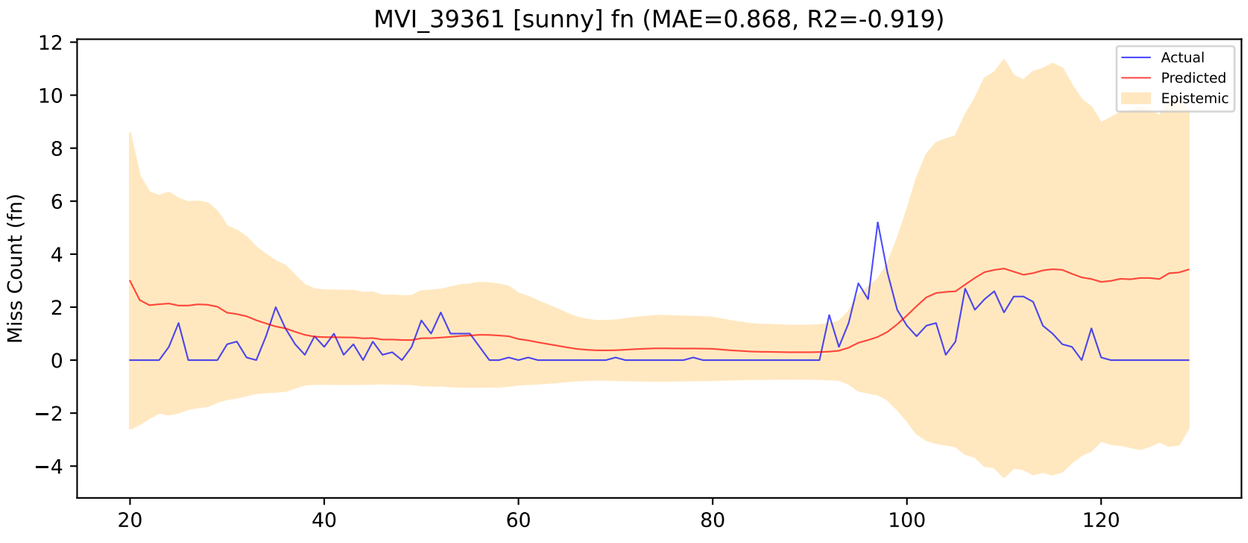
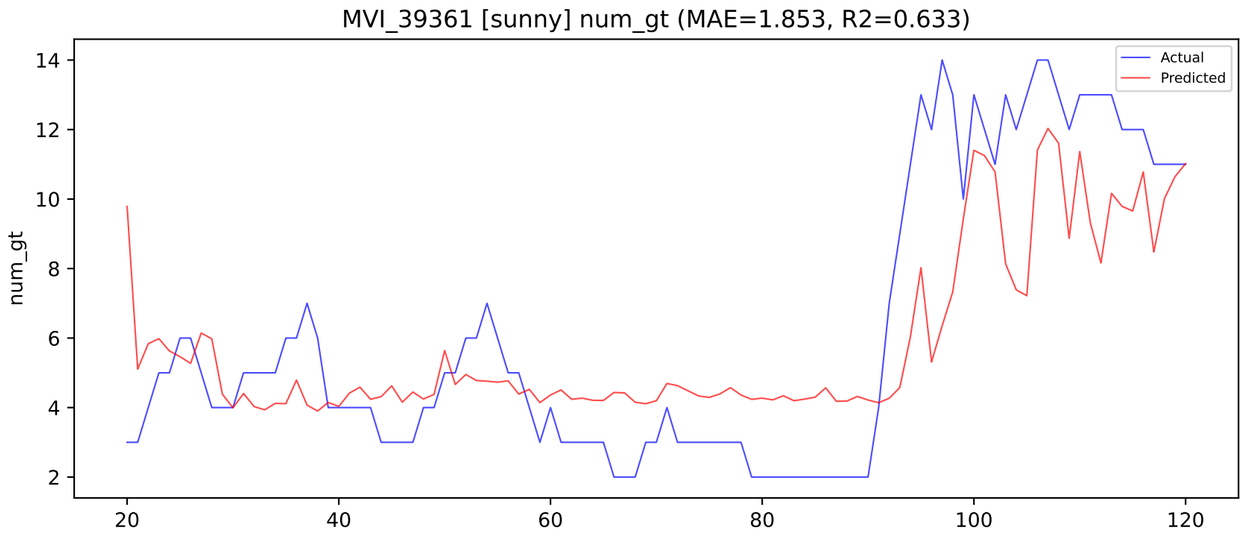
Spatial LSTM (15 features) -- Miss Count Prediction -- Top 2 Sequences per Weather

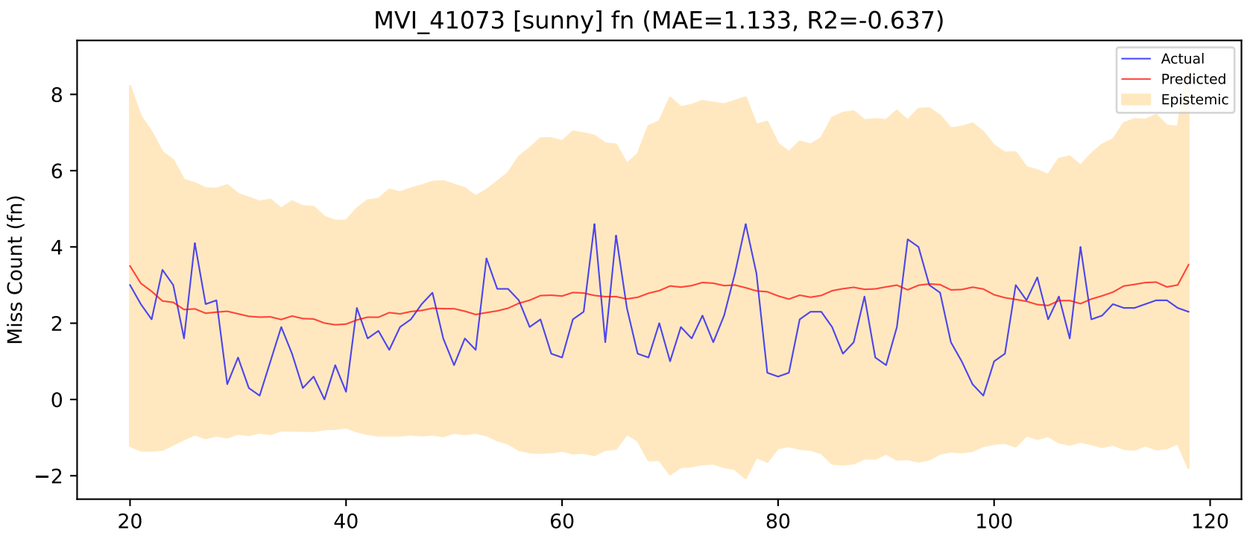
MVI_39361 [sunny] fn (MAE=0.868, R2=-0.919)



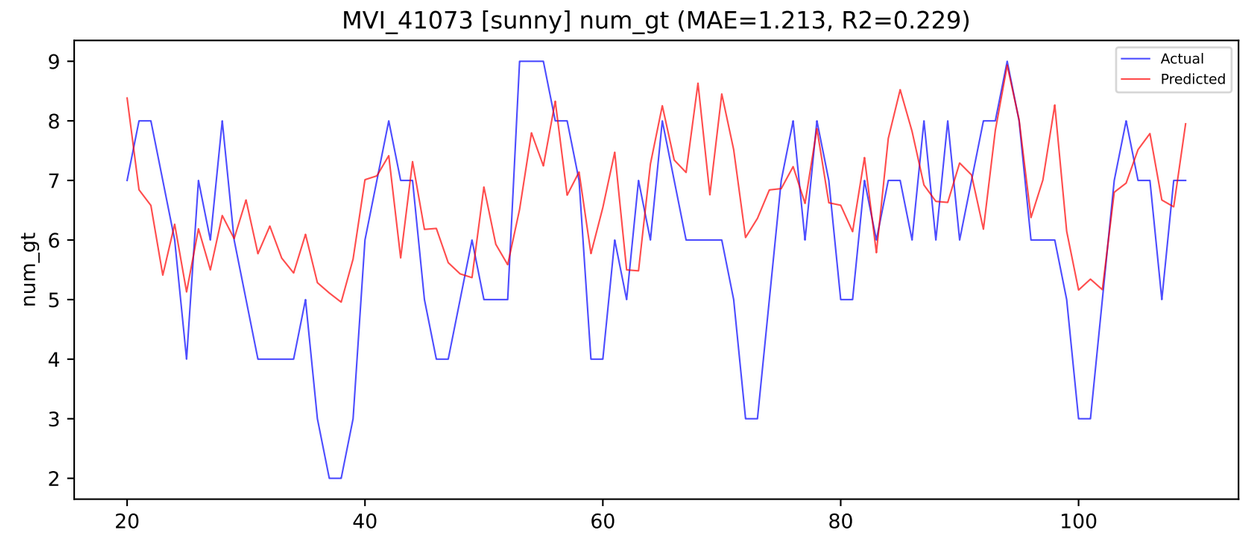
MVI_39361 [sunny] num_gt (MAE=1.853, R2=0.633)



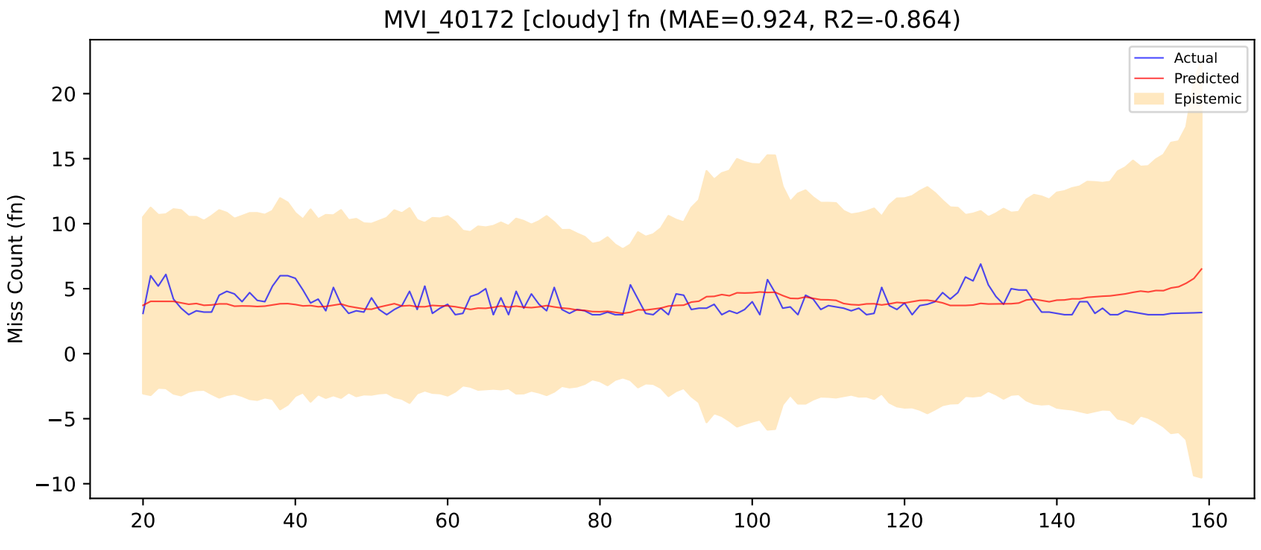
MVI_41073 [sunny] fn (MAE=1.133, R2=-0.637)



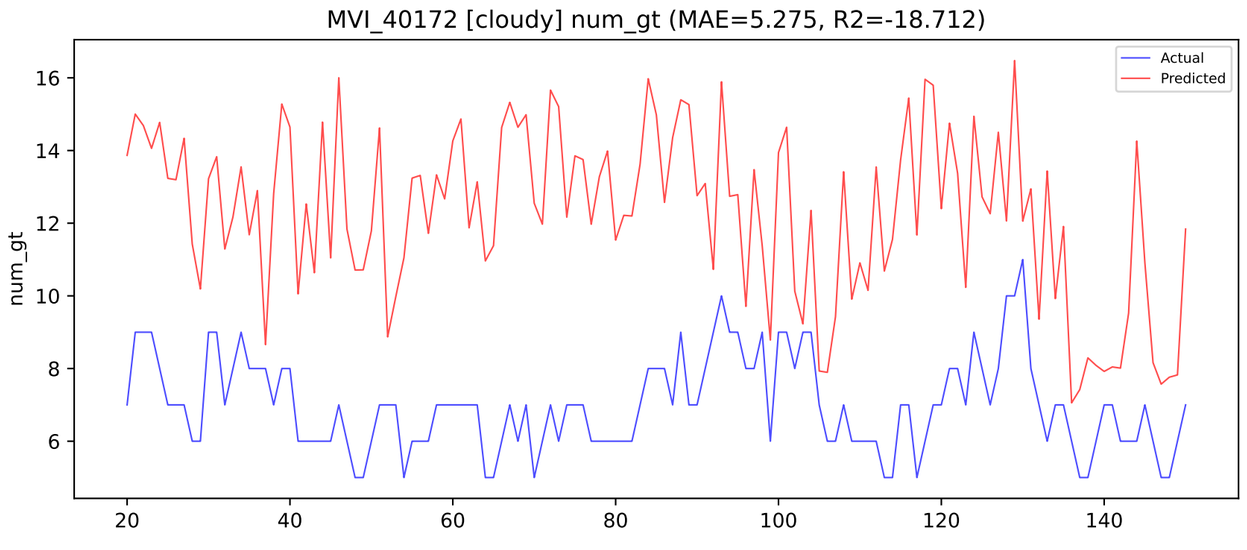
MVI_41073 [sunny] num_gt (MAE=1.213, R2=0.229)



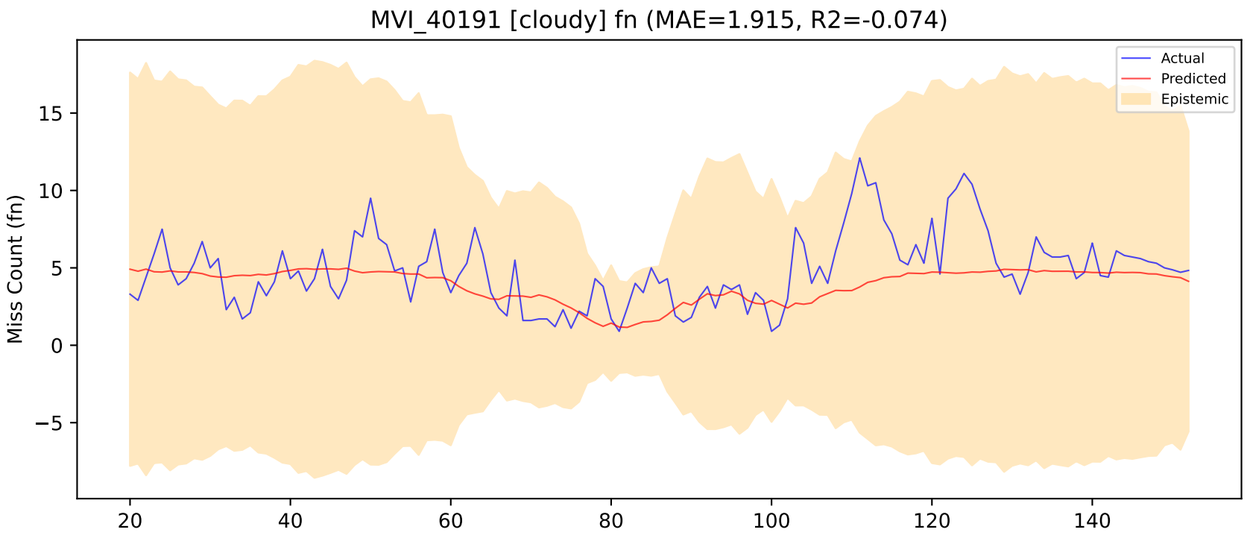
MVI_40172 [cloudy] fn (MAE=0.924, R2=-0.864)



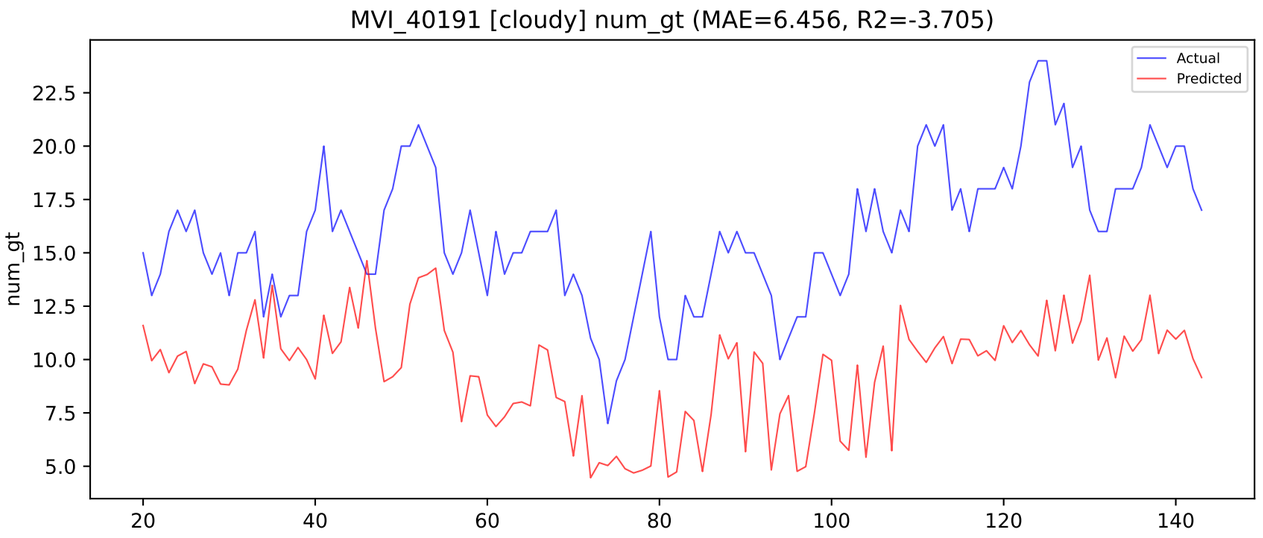
MVI_40172 [cloudy] num_gt (MAE=5.275, R2=-18.712)



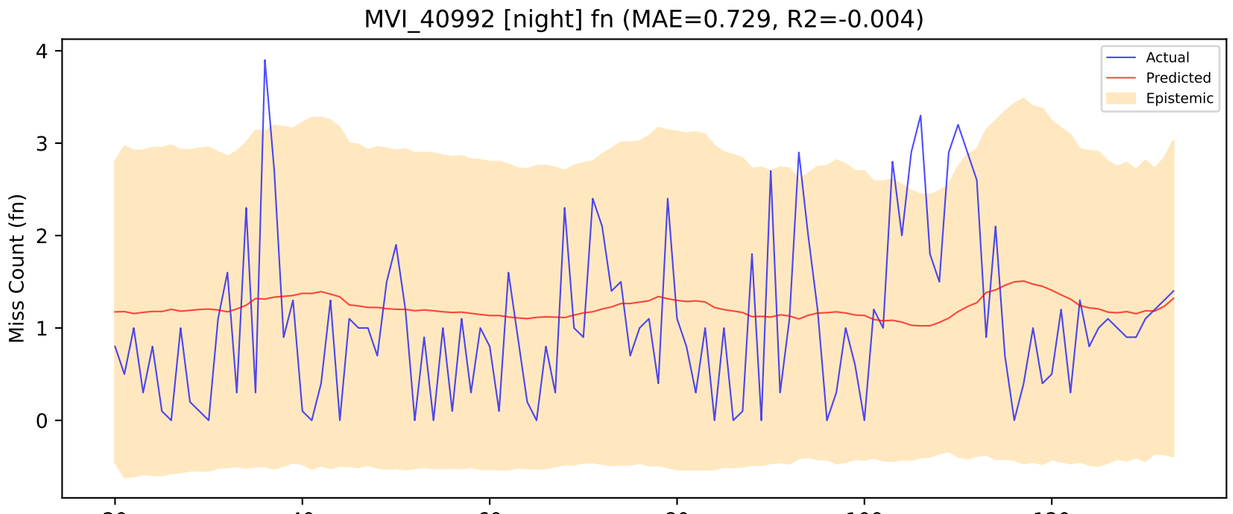
MVI_40191 [cloudy] fn (MAE=1.915, R2=-0.074)



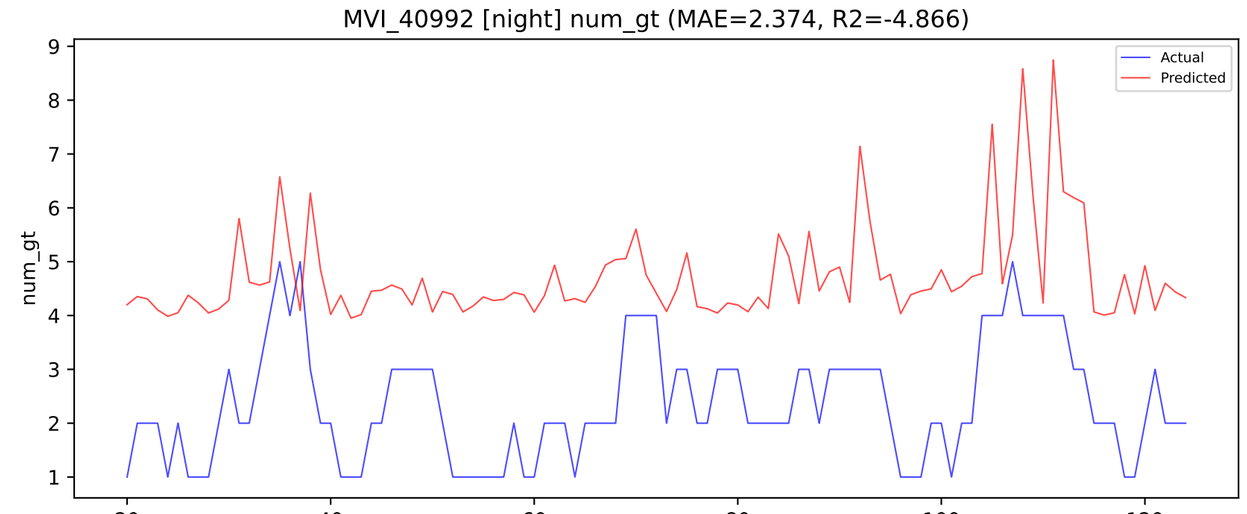
MVI_40191 [cloudy] num_gt (MAE=6.456, R2=-3.705)



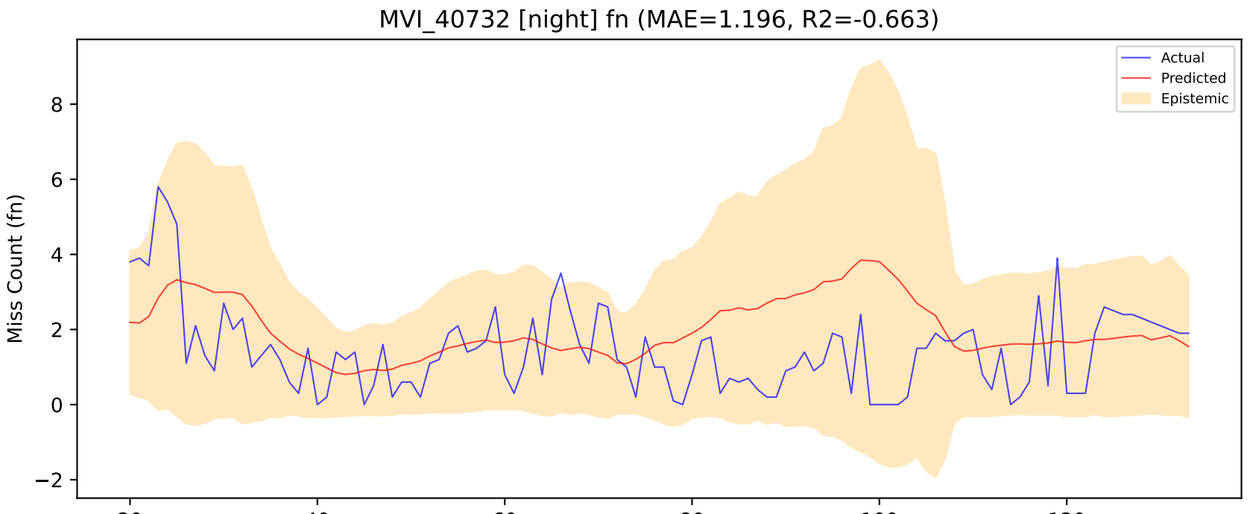
MVI_40992 [night] fn (MAE=0.729, R2=-0.004)



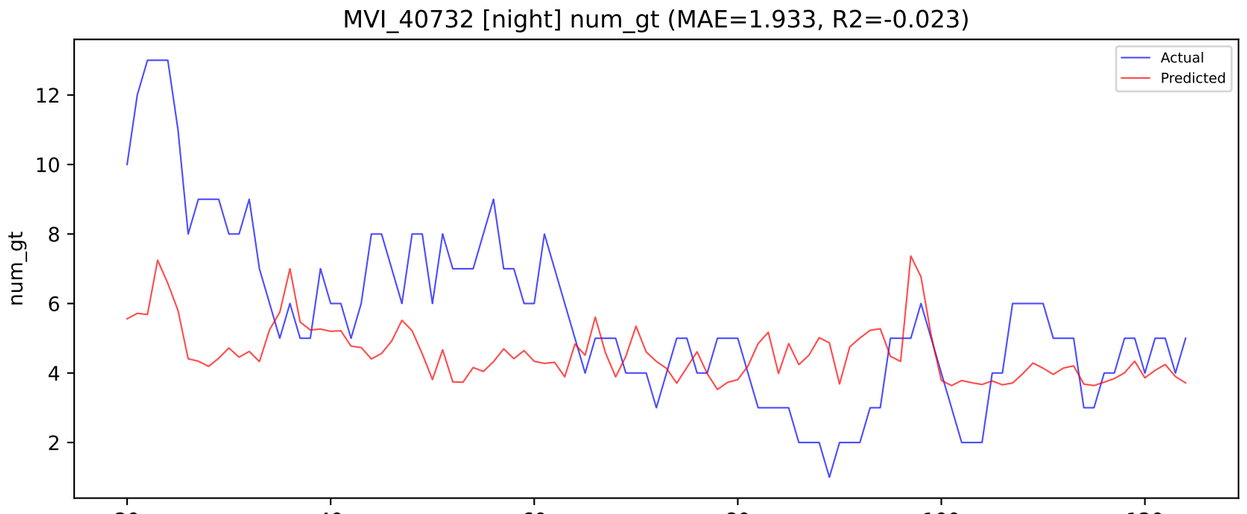
MVI_40992 [night] num_gt (MAE=2.374, R2=-4.866)



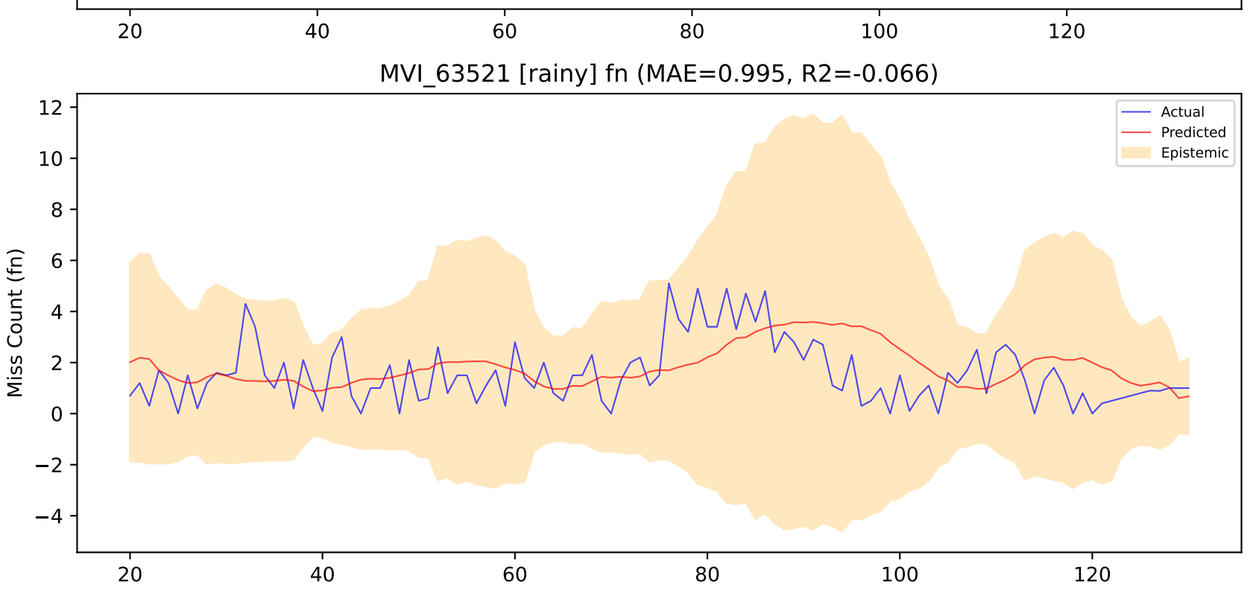
MVI_40732 [night] fn (MAE=1.196, R2=-0.663)



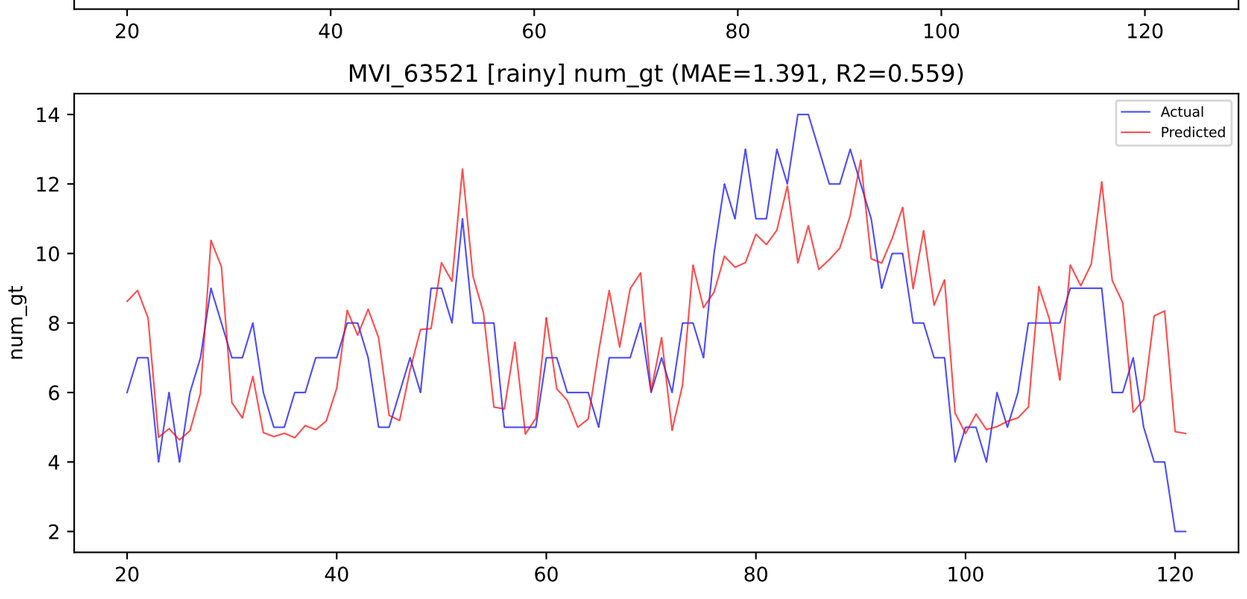
MVI_40732 [night] num_gt (MAE=1.933, R2=-0.023)



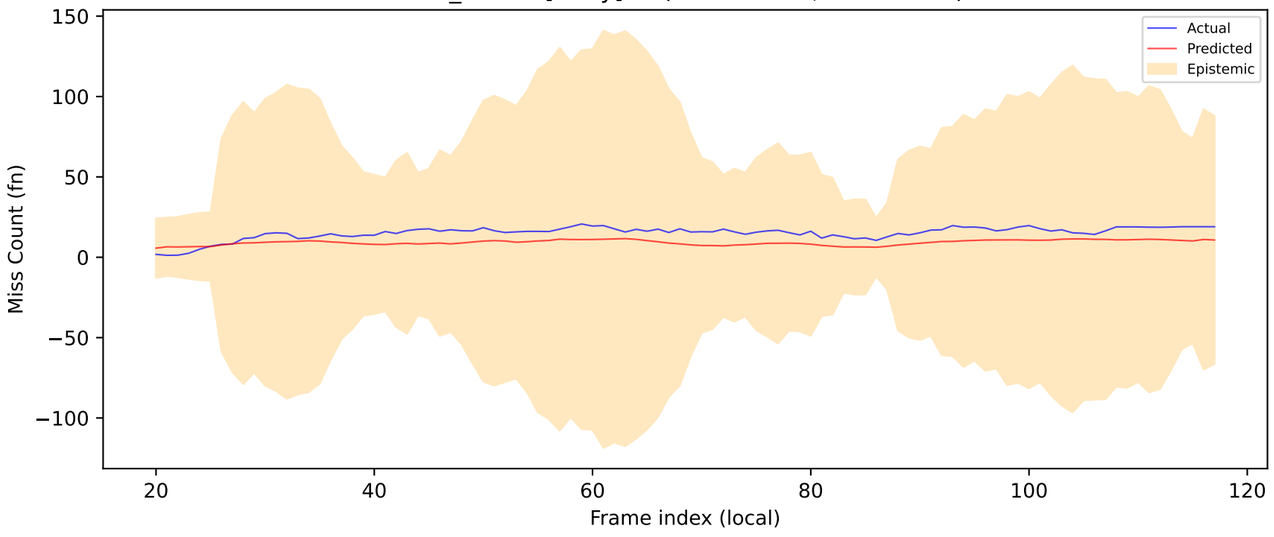
MVI_63521 [rainy] fn (MAE=0.995, R2=-0.066)



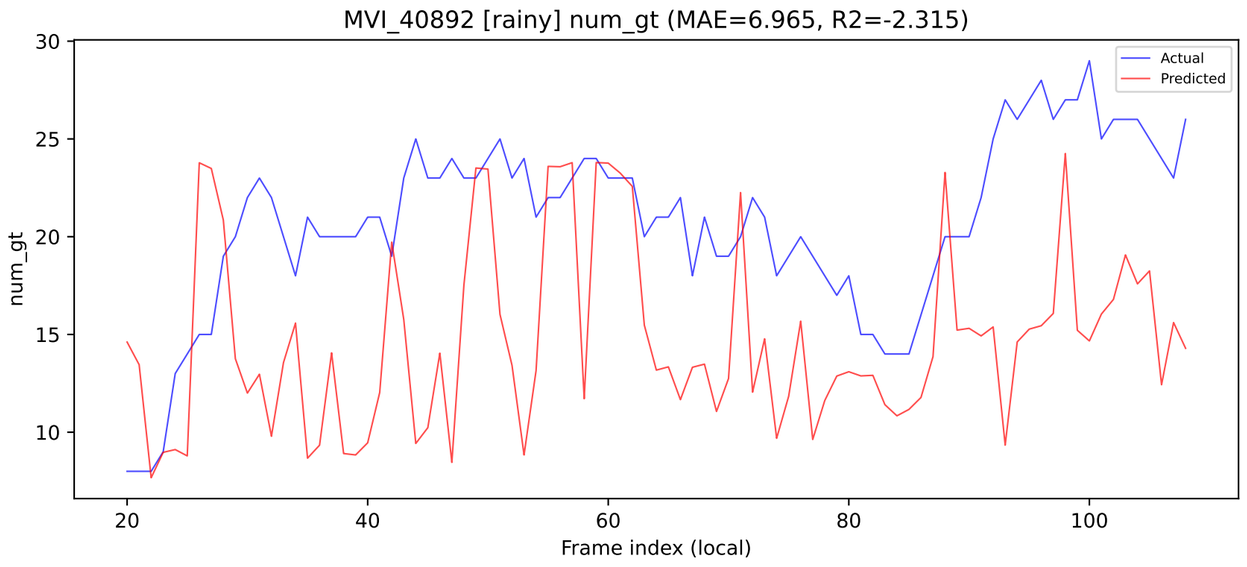
MVI_63521 [rainy] num_gt (MAE=1.391, R2=0.559)



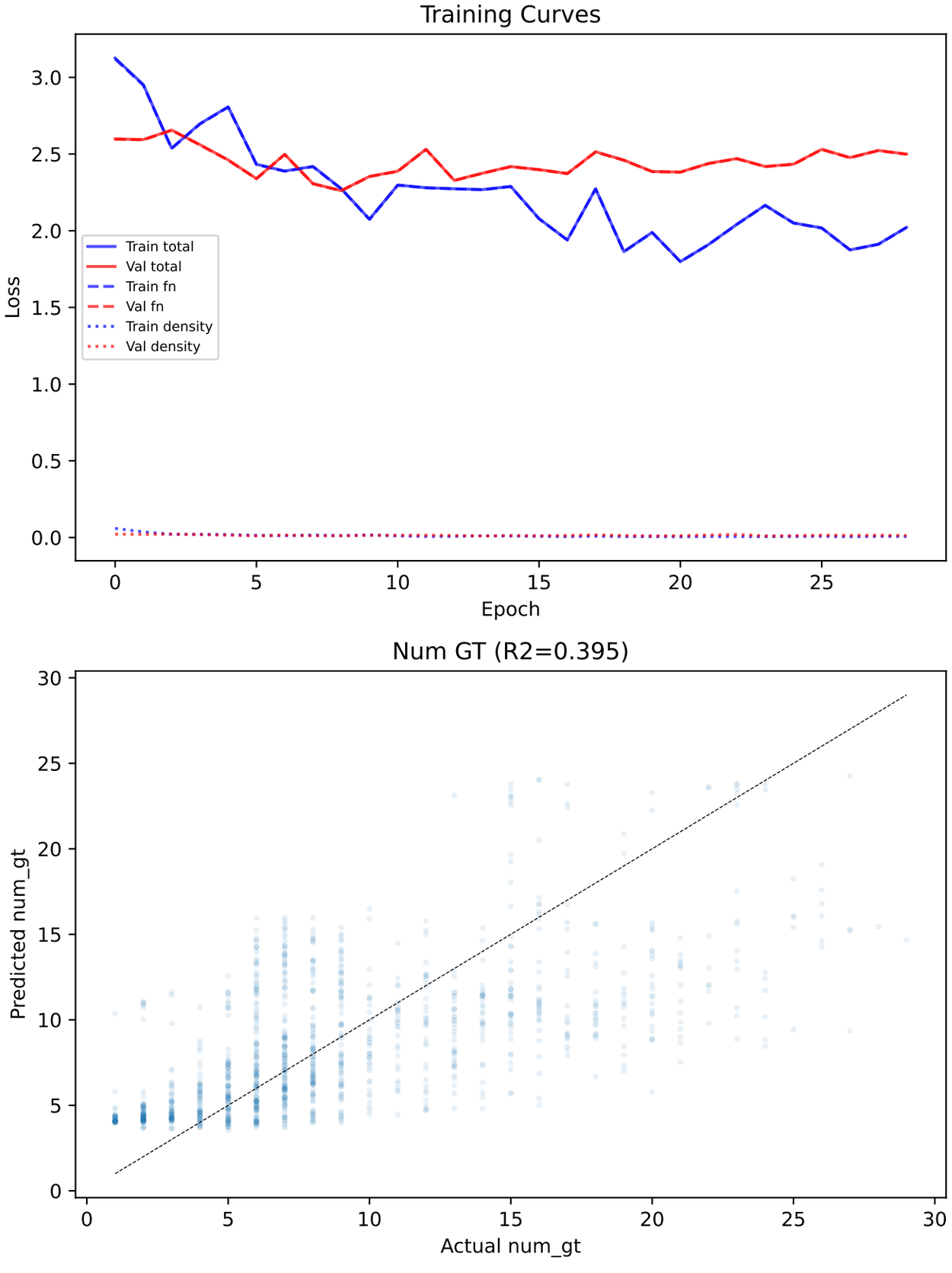
MVI_40892 [rainy] fn (MAE=6.313, R2=-2.097)

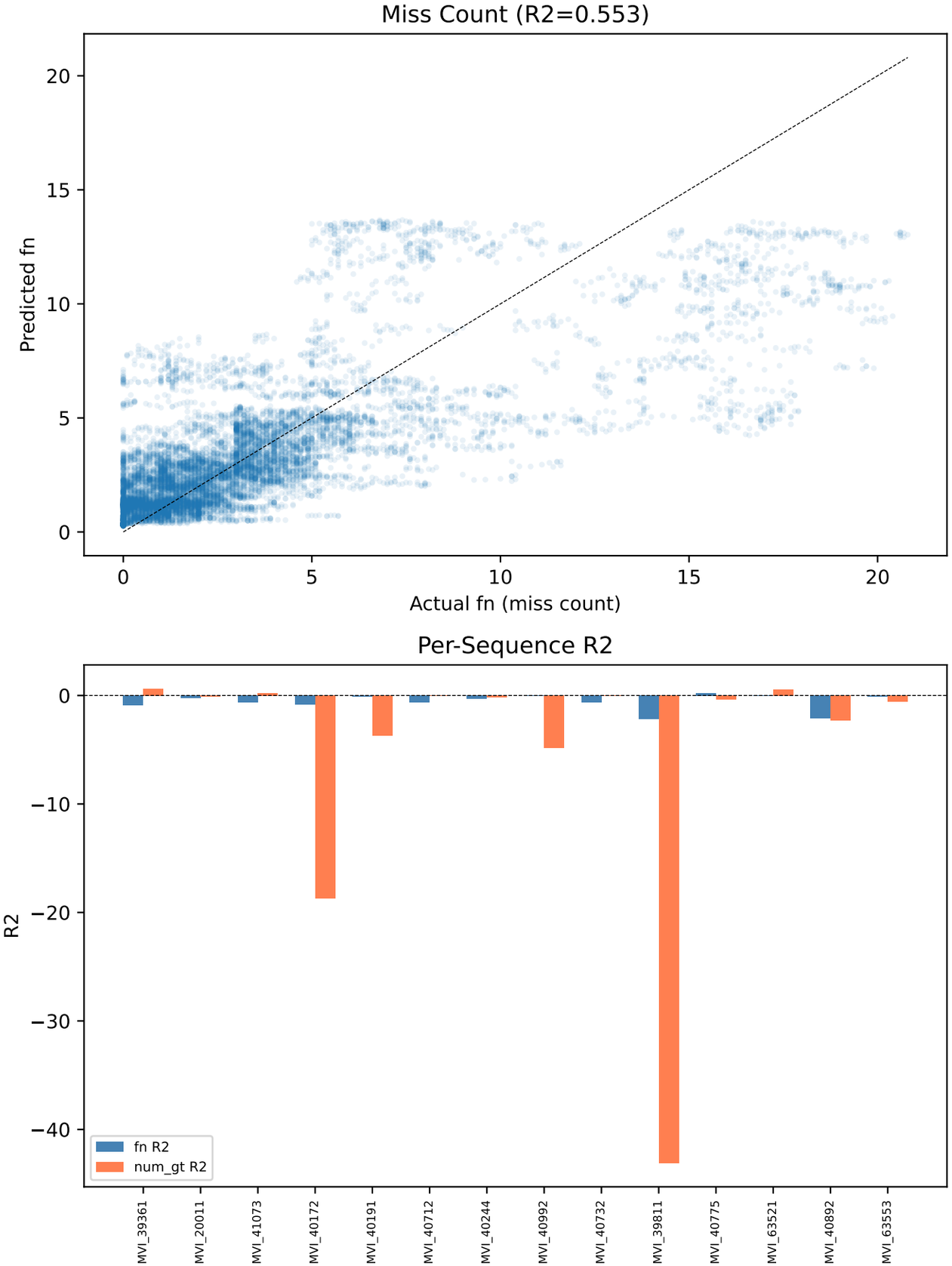


MVI_40892 [rainy] num_gt (MAE=6.965, R2=-2.315)



Page 2: Diagnostics





Page 3: Per-Sequence Metrics

Spatial LSTM -- Validation Metrics (15 spatial features, target=fn)

Sequence	Weather	fn MAE	fn R2	fn rho	GT MAE	GT R2	GT rho
MVI_39361	sunny	0.868	-0.9192	0.6634	1.853	0.6332	0.6804
MVI_20011	sunny	1.778	-0.2620	-0.3835	2.392	-0.0897	0.3231
MVI_41073	sunny	1.133	-0.6367	0.1072	1.213	0.2292	0.6094
MVI_40172	cloudy	0.924	-0.8641	-0.0351	5.275	-18.7122	0.4175
MVI_40191	cloudy	1.915	-0.0738	0.3300	6.456	-3.7054	0.6036
MVI_40712	cloudy	3.248	-0.6383	0.8355	3.497	0.0092	0.7448
MVI_40244	cloudy	1.010	-0.3291	-0.0029	1.851	-0.1348	0.1714
MVI_40992	night	0.729	-0.0038	0.0532	2.374	-4.8665	0.4917
MVI_40732	night	1.196	-0.6632	0.0971	1.933	-0.0228	0.2851
MVI_39811	night	0.728	-2.1958	-0.3154	2.841	-43.0981	-0.0695
MVI_40775	night	2.021	0.2034	0.7379	5.466	-0.3523	0.7606
MVI_63521	rainy	0.995	-0.0659	0.3370	1.391	0.5591	0.8119
MVI_40892	rainy	6.313	-2.0972	0.2712	6.965	-2.3154	0.3465
MVI_63553	rainy	0.961	-0.1220	-0.0132	1.926	-0.5567	0.3557
OVERALL		1.746	0.5527	0.7311	3.386	0.3949	0.6833
BASELINE	(mean)	2.781	0.0000	nan	4.558	0.0000	nan

Page 4: Cross-Experiment Comparison

Cross-Experiment Comparison (all LSTM experiments on DETRAC)

Experiment	Target	R2	MAE	rho
Image-only (14 feat)	recall_gap	-0.1150	0.1440	0.0840
Both models (32 feat)	recall_gap	0.1650	0.1230	0.5300
LSTM_nano (23 feat)	recall_gap	-0.0170	0.1370	0.3570
LSTM_medium (23 feat)	recall_gap	-0.0980	0.1440	0.2150
Nano recall (23 feat)	nano_recall	-0.0210	0.1910	0.2650
Spatial (15 feat)	miss_count (fn)	0.5527	1.7464	0.7311

